# MACHINE LEARNING PROJECT REPORT

# COLLEGE ADMISSION CHANCE PREDICTION

Using Regression Algorithms and Student Academic Features

**Project Members**
HETVI RAJODIYA
KHUSHI PATEL

**Roll Numbers**
504 505

**Subject**
Machine Learning

**Academic Year**
2026–2027

Submitted as a Machine Learning Project Report

# 1. Certificate

This is to certify that the project report entitled “College Admission Chance Prediction” has been prepared by HETVI RAJODIYA (Roll No. 504) and KHUSHI PATEL (Roll No. 505) as part of the Machine Learning subject. The report documents the dataset, preprocessing, exploratory analysis, regression model development, evaluation, results, conclusion, and future scope.

| Student Signature | Faculty Signature | Date |
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## 2. Problem Statement

Choosing a college and estimating admission prospects can be difficult for students. Admission decisions may depend on academic performance, standardized test scores, university profile, recommendation ratings, statement of purpose, research exposure, and experience. This project develops a supervised machine learning regression model that estimates an illustrative admission-chance score from these inputs.

The output is a continuous score between 0 and 1. It is an educational estimate generated from a synthetic dataset and must not be treated as an official probability or a guarantee of admission.

## 3. Objectives

• Create and analyse a structured dataset containing 2,500 student records and 10 fields.

• Inspect data types, missing values, duplicate rows, and descriptive statistics.

• Explore relationships between student attributes and admission chance using graphs.

• Split the data into training and testing sets using an 80:20 ratio.

• Train Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor models.

• Compare models using MAE, RMSE, and R-squared metrics.

• Save the best-performing model and use it to predict a sample student's admission-chance score.

## 4. Dataset Description

The dataset is stored as college_admission_2500_10_fields.csv. It contains 2,500 records and 10 fields. The dataset is synthetically generated for academic demonstration; it does not contain real student records or represent a particular college's admission policy.

| Field | Description |
|---|---|
| GRE_Score | GRE score (260–340) |
| TOEFL_Score | TOEFL score (75–120) |
| University_Rating | University rating on a 1–5 scale |
| SOP_Rating | Statement of Purpose rating on a 1–5 scale |
| LOR_Rating | Letter of Recommendation rating on a 1–5 scale |
| CGPA | Academic grade point average (5.50–10.00) |
| Research | Research experience indicator (0 = No, 1 = Yes) |
| Work_Experience | Years of work experience (0–5) |
| Internship_Count | Number of internships (0–4) |
| Admission_Chance | Target score between 0.05 and 0.99 |

**Dataset shape:** 2500 rows × 10 columns. **Duplicate records:** 0. **Missing values:** 0.

## 5. Data Preprocessing

The dataset is loaded using Pandas. Duplicate rows are removed and incomplete rows are excluded before separating the input features and target variable. All nine student attributes are used as input features, while Admission_Chance is the continuous target.

• Load the CSV dataset into a Pandas DataFrame.

• Inspect the first records, column types, missing values, and duplicate rows.

• Remove duplicate rows and handle missing records.

• Separate the nine input variables (X) from Admission_Chance (y).

• Split the dataset into 80% training records and 20% testing records using random_state=42.

## 6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to understand the target distribution and relationships between student attributes. The following charts are generated from the dataset.

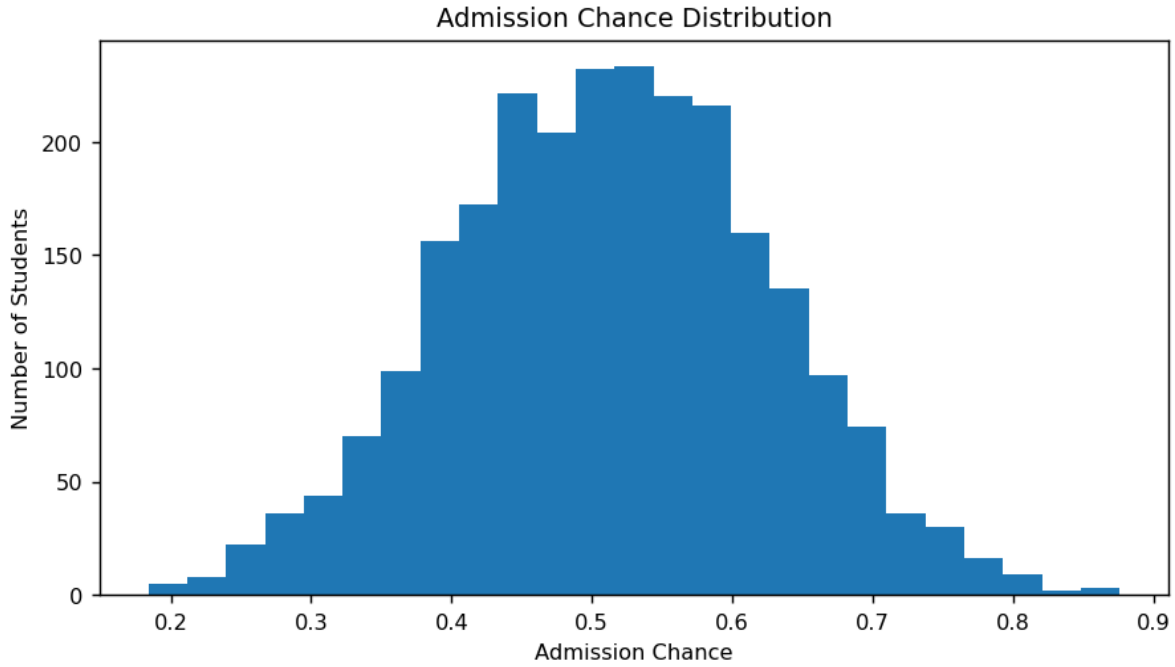



Figure 1. Distribution of admission-chance scores.

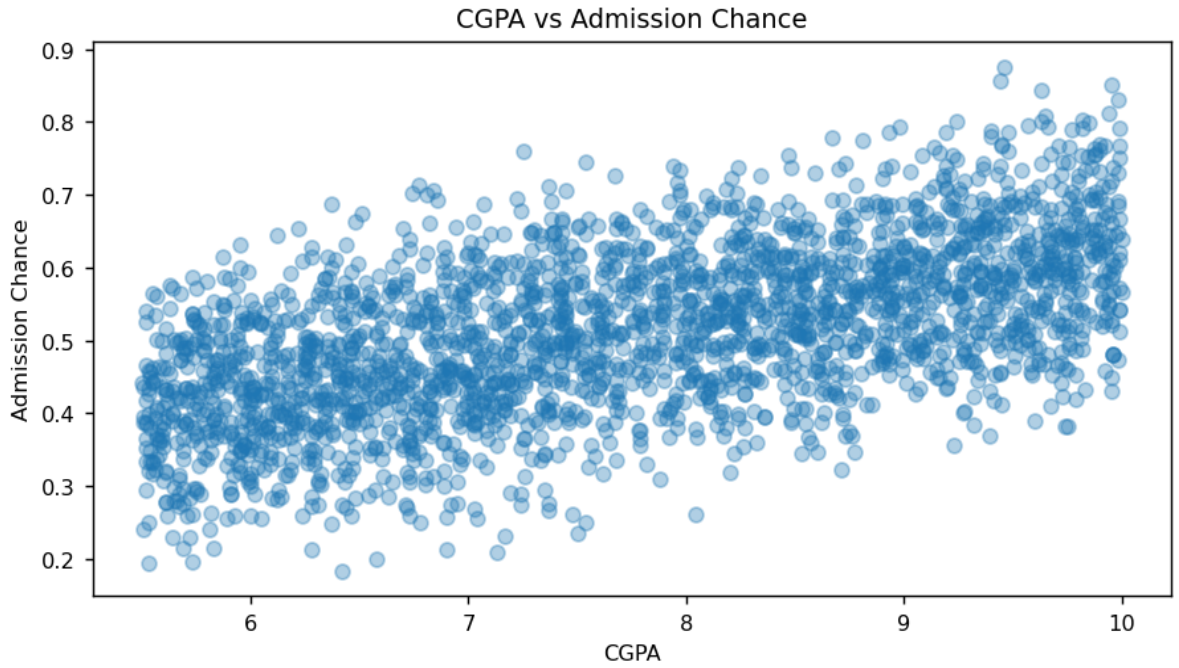



Figure 2. Relationship between CGPA and admission chance.

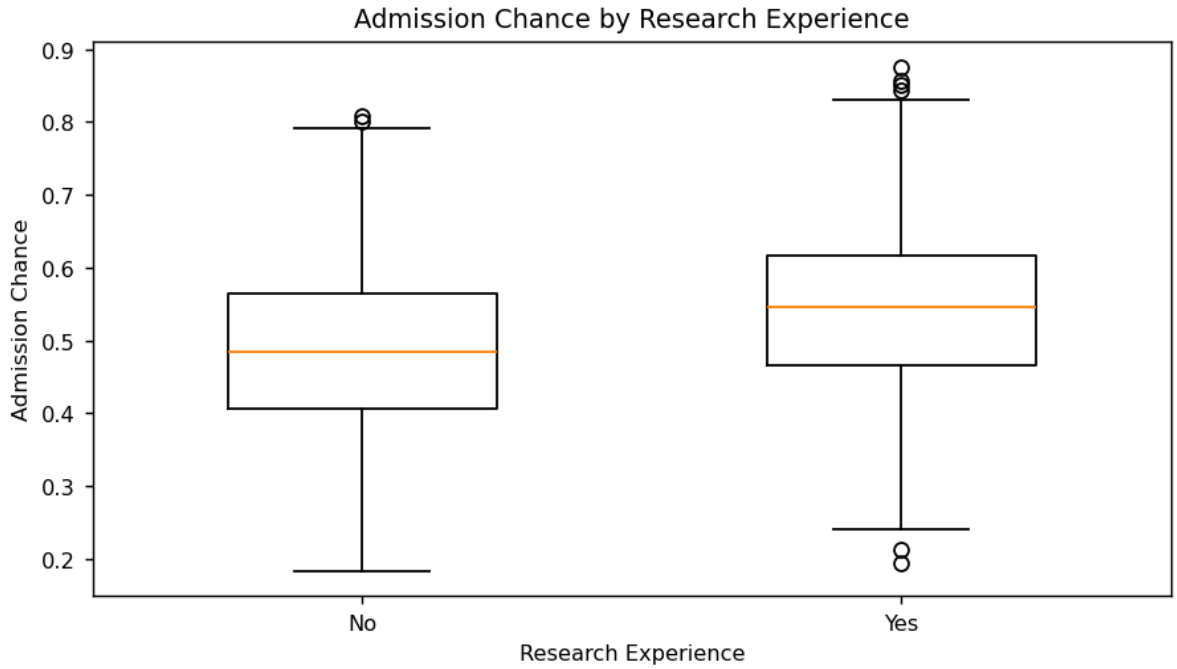



Figure 3. Admission chance grouped by research experience.

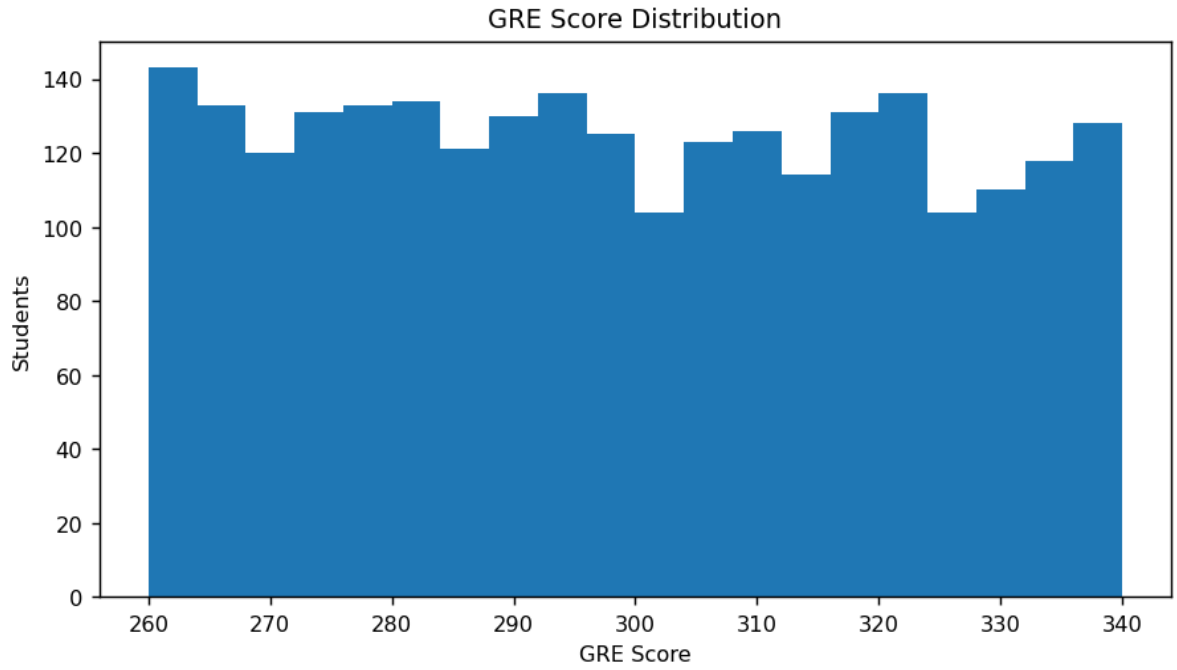



Figure 4. Distribution of GRE scores.

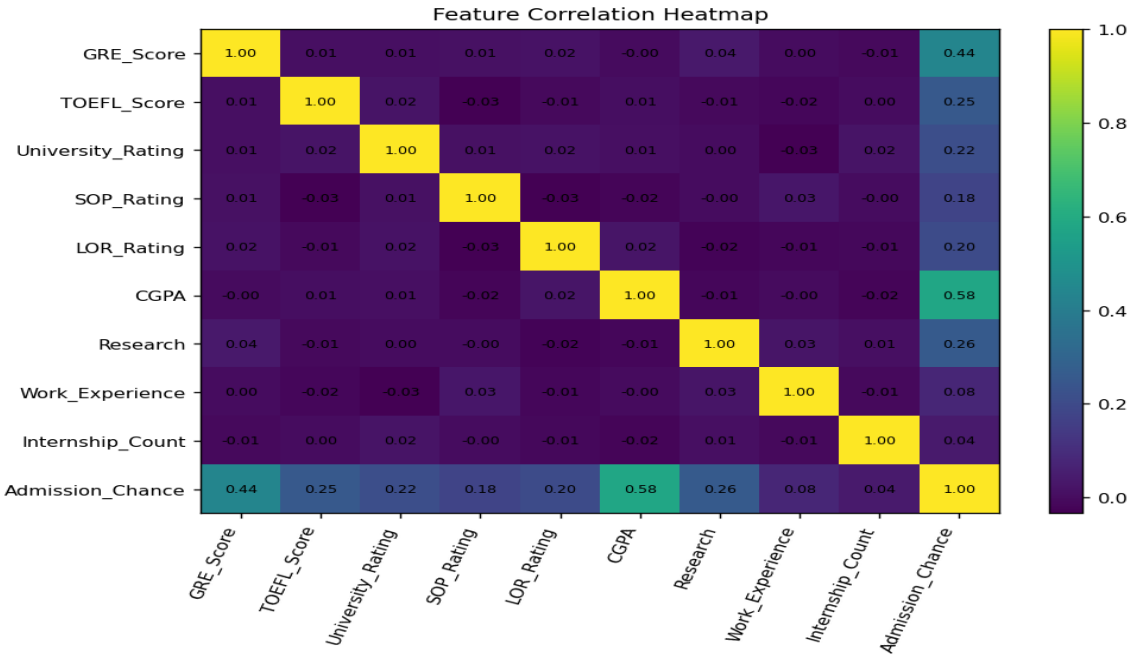



Figure 5. Correlation heatmap for numerical fields.

The visual analysis provides a quick overview of score ranges, the relationship between CGPA and the target, differences by research experience, and correlations among the numerical variables.

## 7. Model Building

This is a regression task because Admission_Chance is a continuous numeric value. Three regression algorithms are trained and compared:

• Linear Regression: a baseline model that estimates a linear relationship between the input features and target. StandardScaler is used within a pipeline.

• Random Forest Regressor: combines multiple decision trees and averages their predictions.

• Gradient Boosting Regressor: builds an ensemble of trees sequentially to reduce prediction error.

**Training/testing:** 2,000 records are used for training and 500 records for testing (80:20 split).

**Model selection:** The model with the lowest RMSE on the held-out test set is selected as the final model.

## 8. Model Evaluation

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| Linear Regression | 0.0431 | 0.0535 | 0.7724 |
| Random Forest | 0.0486 | 0.0604 | 0.7098 |
| Gradient Boosting | 0.0466 | 0.0581 | 0.7313 |

The best model on this run is **Linear Regression**, selected because it has the lowest RMSE (0.0535). MAE measures the average absolute prediction error. RMSE penalizes larger errors more strongly. R-squared describes the proportion of target variation explained by the model.

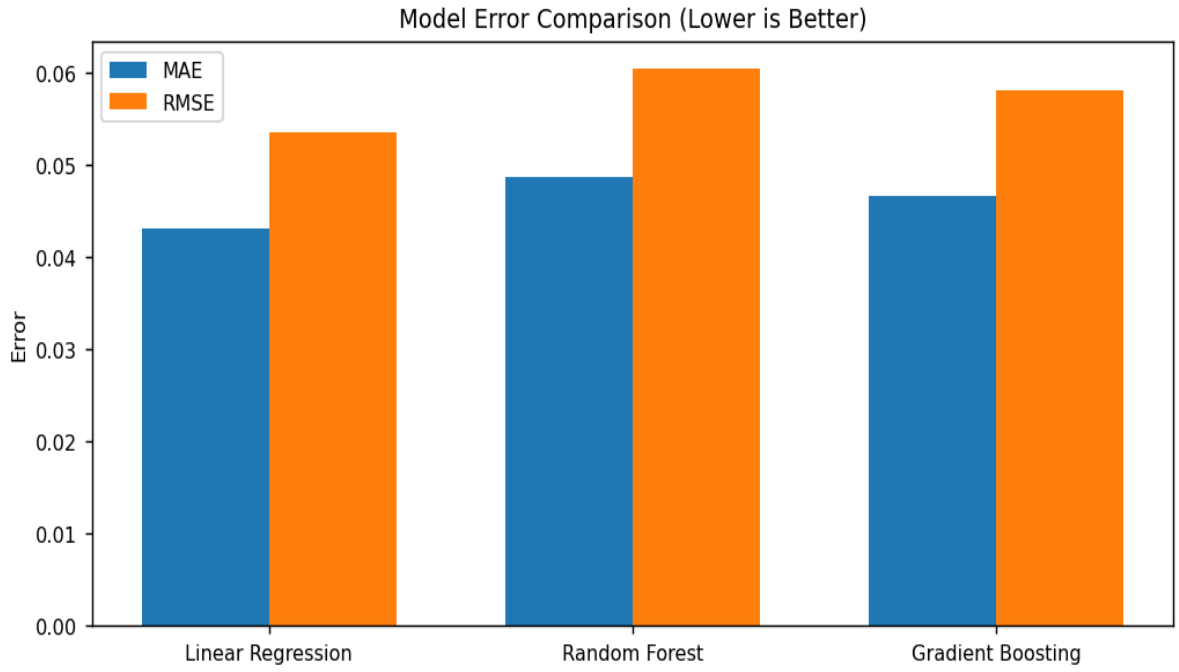



Figure 6. MAE and RMSE comparison (lower values are better).

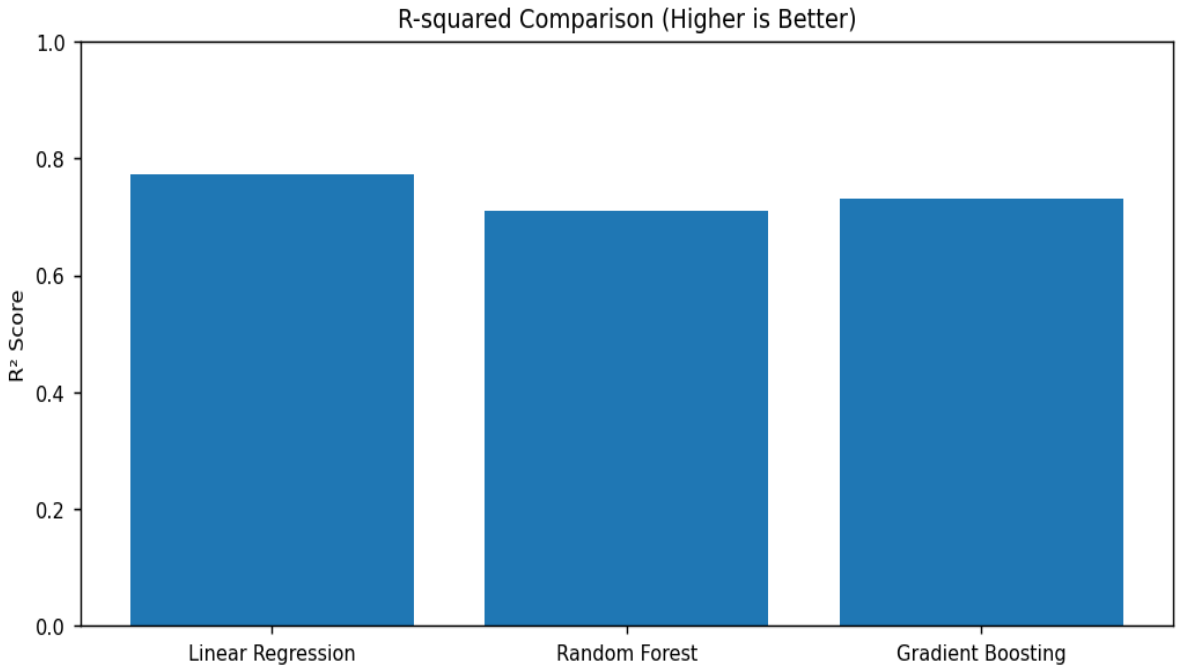



Figure 7. R-squared comparison (higher values are better).

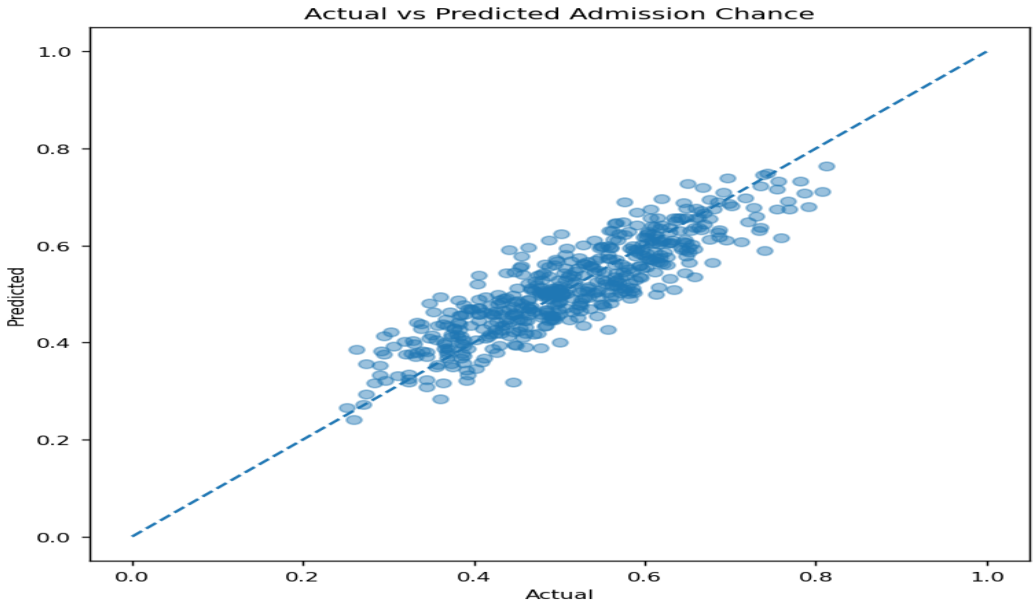



Figure 8. Actual versus predicted values for the selected model.

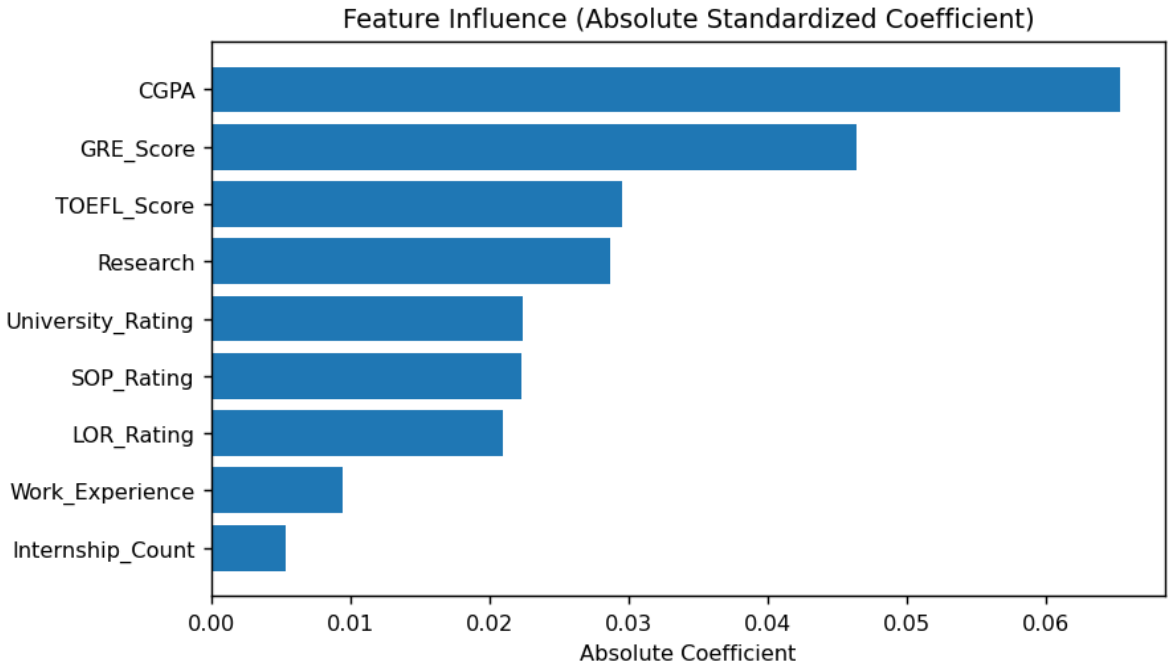



Figure 9. Absolute standardized coefficient sizes for the Linear Regression model. This is a descriptive measure of model coefficients, not proof of causation.

## 9. Results

On the 500-record test set, the selected Linear Regression model achieved:

• Mean Absolute Error (MAE): 0.0431

• Root Mean Squared Error (RMSE): 0.0535

• R-squared ($R^2$): 0.7724

These values describe performance on this synthetic dataset and fixed train/test split. Results may differ if the dataset, random seed, or model settings change.

## 10. Sample Prediction

| Input field | Sample value |
|---|---|
| GRE Score | 315 |
| TOEFL Score | 105 |
| University Rating | 4 |
| SOP Rating | 4.0 |
| LOR Rating | 4.0 |
| CGPA | 8.50 |
| Research | Yes |
| Work Experience | 2 years |
| Internship Count | 2 |
| Predicted Admission Chance | 0.682 (68.2%) |

The sample prediction is illustrative only. It should not be interpreted as a real institution's admission decision or an objectively calibrated probability.

## 11. Conclusion

The project demonstrates an end-to-end machine learning regression workflow for estimating a college admission-chance score. A synthetic dataset with 2,500 records and 10 fields was explored, prepared, and divided into training and testing sets. Three regression algorithms were compared using MAE, RMSE, and R-squared. The best model was saved so it can be reused for new sample inputs.

## 12. Future Scope

• Use a verified real-world dataset with documented consent and source information.

• Add relevant factors such as program, country, application year, and institution-specific requirements.

• Evaluate the model using cross-validation and a separate external test dataset.

• Check fairness, data quality, and prediction calibration before any real-world use.

• Build a simple web interface for entering student details and viewing the estimate.

## 13. Project Files

• Dataset/college_admission_2500_10_fields.csv — 2,500-row dataset.

• python_code/all code.py — complete Python implementation.

• Images of model/ — EDA and evaluation graphs.

• Model/ — saved trained model, feature list, and model comparison CSV.

• Reports/EDA_Report_College_Admission.html — HTML EDA report.

**Academic note:** This project is intended for learning and demonstration. The synthetic values and model outputs are not validated for actual admissions decisions.